



6590 - A New Sample of Dwarf Galaxies with Broad-line AGN

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) GAMA718498
	2		NIRCam Imaging	(2) GAMA2123678
	3		NIRCam Imaging	(3) GAMA5240292
	4		NIRCam Imaging	(4) GAMA98560
	5		NIRCam Imaging	(5) GAMA2258819
	6		NIRCam Imaging	(6) GAMA593724
	7		NIRCam Imaging	(7) GAMA31941
	8		NIRCam Imaging	(8) GAMA5275222
	9		NIRCam Imaging	(9) GAMA484908
	10		NIRCam Imaging	(10) GAMA238411

ABSTRACT

The study of massive black holes (BHs) in dwarf galaxies can place valuable constraints on the birth and growth of the first supermassive BH seeds. We will obtain Chandra and JWST observations of dwarf galaxies from a recent sample of optically-selected AGN in low-mass galaxies from the Galaxy And Mass Assembly (GAMA) Survey. Our target galaxies have stellar masses less than 5 billion Msun and host broad-line AGNs with spectroscopic BH mass estimates. The proposed observations will enable us to characterize the compact X-ray emission from the AGNs, estimate

Eddington ratios, and investigate host galaxy properties and BH scaling relations at the sparsely-populated low-mass end.
(Chandra Proposal Number 26700402)

OBSERVING DESCRIPTION

As part of our Chandra proposal, we also propose joint infrared observations with NIRCAM aboard JWST in one short/long wavelength filter pair (F200W and F444W). The proposed 2 m (F200W; K-band) imaging will enable us to determine the detailed morphologies and structural components of the galaxies with sufficient resolution and sensitivity to detect small, faint bulges, as well as investigate BH scaling relations such as MBH Lbulge and MBH Mbulge at the sparsely-populated low-mass end. With X-ray data from the proposed Chandra observations, we will also be able to look for trends in (or offsets from) these relations with Eddington ratio. As a bonus, we will obtain simultaneous imaging at 4.4 m (F444W), which will detect and disentangle potential AGN-heated dust from stellar/star-formation-related emission from the host galaxy. In combination with our proposed Chandra observations, these data will allow us to calculate the X-ray to mid-IR ratio for our target AGNs and compare to established relations for more massive galaxies hosting AGNs. Our science goals can only be achieved with the superb angular resolution and sensitivity provided by JWST/NIRCAM at infrared wavelengths. Ultimately, the combination of Chandra and JWST observations will provide crucial insights into BH growth in dwarf galaxies.

Instrumental Setup:

All our target galaxies have angular sizes < 1 arcmin and will fit on a single short wavelength detector. We will therefore use a single module (module B) with an offset (40, 40) to center the target on a single short wavelength detector (B1) and avoid gaps. The full array will be used to completely cover the galaxy with both the short and long wavelength detectors and facilitate sky subtraction. We will use the Standard Subpixel Dither pattern with 4 subpixel positions in each filter (see below) to improve sampling in the stacked images. Primary Dithers are not needed since there will be no gaps in sky coverage for our targets. We assume a medium background level for all our calculations using the Exposure Time Calculator (ETC).

Exposure Times & Readout Patterns:

Based on the magnitudes of our faintest targets, we request a total exposure time of 18 minutes on each source, which will enable us to reach a depth of at least 27.5 AB magnitudes at 10 and detect small, faint bulges in our target galaxies. Following the recommended readout specifications for maximal depth in a given exposure time, we will use the SHALLOW4 readout pattern with 5-6 groups per integration and 1 integration per exposure. With 4 subpixel dither positions (i.e., 4 exposures), this corresponds to a total exposure time of ~17-20 minutes the short + long filter pair.

Proposal 6590 - Targets - A New Sample of Dwarf Galaxies with Broad-line AGN

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	GAMA718498	RA: 12 02 4.8504 (180.5202100d) Dec: +01 50 8.41 (1.83567d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(2)	GAMA2123678	RA: 02 19 13.3104 (34.8054600d) Dec: -05 43 21.25 (-5.72257d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(3)	GAMA5240292	RA: 22 53 4.5096 (343.2687900d) Dec: -33 58 38.10 (-33.97725d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(4)	GAMA98560	RA: 11 55 5.1288 (178.7713700d) Dec: +00 51 25.31 (.85703d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(5)	GAMA2258819	RA: 02 03 29.7792 (30.8740800d) Dec: -09 29 29.29 (-9.49147d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(6)	GAMA593724	RA: 14 30 13.8096 (217.5575400d) Dec: -00 10 20.57 (-.17238d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(7)	GAMA31941	RA: 12 09 47.4696 (182.4477900d) Dec: -01 04 10.34 (-1.06954d) Equinox: J2000 Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		

Proposal 6590 - Targets - A New Sample of Dwarf Galaxies with Broad-line AGN

(8)	GAMA5275222	RA: 23 01 25.2288 (345.3551200d) Dec: -34 12 40.90 (-34.21136d) Equinox: J2000
Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(9)	GAMA484908	RA: 14 12 58.4208 (213.2434200d) Dec: -01 48 23.36 (-1.80649d) Equinox: J2000
Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		
(10)	GAMA238411	RA: 14 16 59.3808 (214.2474200d) Dec: +01 34 26.33 (1.57398d) Equinox: J2000
Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES		

Proposal 6590 - Observation 1 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 1 Diagnostic Status: Warning Observing Template: NIRCcam Imaging										Mon Jun 02 23:00:08 GMT 2025
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(1)	GAMA718498	RA: 12 02 4.8504 (180.5202100d) Dec: +01 50 8.41 (1.83567d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES										
	Module					Subarray					
Dithers	B					FULL					
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	NONE				STANDARD				4	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
Special Requirements	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73		
	Offset 40.0 arcsec, 40.0 arcsec										

Proposal 6590 - Observation 2 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 2 Diagnostic Status: Warning Observing Template: NIRCam Imaging										Mon Jun 02 23:00:08 GMT 2025
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(2)	GAMA2123678	RA: 02 19 13.3104 (34.8054600d) Dec: -05 43 21.25 (-5.72257d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F444W	SHALLOW4	6	1	4	4	1245.465		
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec										

Proposal 6590 - Observation 3 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 3 Diagnostic Status: Warning Observing Template: NIRCcam Imaging										Mon Jun 02 23:00:08 GMT 2025
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(3)	GAMA5240292	RA: 22 53 4.5096 (343.2687900d) Dec: -33 58 38.10 (-33.97725d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73		
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec										

Proposal 6590 - Observation 4 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 4 Mon Jun 02 23:00:08 GMT 2025									
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging									
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(4)	GAMA98560	RA: 11 55 5.1288 (178.7713700d) Dec: +00 51 25.31 (.85703d) Equinox: J2000							
Template	<i>Comments:</i> Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES									
	Module					Subarray				
Dithers	B									
	FULL									
Spectral Elements	#	Primary Dither Type	Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE			STANDARD				4	
Special Requirements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73	
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec									

Proposal 6590 - Observation 5 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 5 Mon Jun 02 23:00:08 GMT 2025									
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging									
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(5)	GAMA2258819	RA: 02 03 29.7792 (30.8740800d) Dec: -09 29 29.29 (-9.49147d) Equinox: J2000							
Template	<i>Comments:</i> Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES									
	Module					Subarray				
Dithers	B									
	FULL									
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				4
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73	
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec									

Proposal 6590 - Observation 6 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 6 Mon Jun 02 23:00:08 GMT 2025									
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging									
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(6)	GAMA593724	RA: 14 30 13.8096 (217.5575400d) Dec: -00 10 20.57 (-.17238d) Equinox: J2000							
Template	<i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies]</i> <i>Extended=YES</i>									
	Module					Subarray				
Dithers	B									
	FULL									
Spectral Elements	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				STANDARD				4
Special Requirements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73	
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec									

Proposal 6590 - Observation 7 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 7 Diagnostic Status: Warning Observing Template: NIRCam Imaging										Mon Jun 02 23:00:08 GMT 2025
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(7)	GAMA31941	RA: 12 09 47.4696 (182.4477900d) Dec: -01 04 10.34 (-1.06954d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F444W	SHALLOW4	6	1	4	4	1245.465		
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec										

Proposal 6590 - Observation 8 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 8 Diagnostic Status: Warning Observing Template: NIRCam Imaging										Mon Jun 02 23:00:08 GMT 2025
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(8)	GAMA5275222	RA: 23 01 25.2288 (345.3551200d) Dec: -34 12 40.90 (-34.21136d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES										
	Module					Subarray					
Dithers	B										FULL
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	NONE				STANDARD				4	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
Special Requirements	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73		
	Offset 40.0 arcsec, 40.0 arcsec										

Proposal 6590 - Observation 9 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 9 Diagnostic Status: Warning Observing Template: NIRCam Imaging										Mon Jun 02 23:00:08 GMT 2025
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(9)	GAMA484908	RA: 14 12 58.4208 (213.2434200d) Dec: -01 48 23.36 (-1.80649d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES										
	Module					Subarray					
Dithers	B					FULL					
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	NONE				STANDARD				4	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
Special Requirements	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73		
	Offset 40.0 arcsec, 40.0 arcsec										

Proposal 6590 - Observation 10 - A New Sample of Dwarf Galaxies with Broad-line AGN

Observation	Proposal 6590, Observation 10 Mon Jun 02 23:00:08 GMT 2025									
	Diagnostic Status: Warning Observing Template: NIRCam Imaging									
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(10)	GAMA238411	RA: 14 16 59.3808 (214.2474200d) Dec: +01 34 26.33 (1.57398d) Equinox: J2000							
Template	<i>Comments:</i> Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Dwarf galaxies, Emission line galaxies, Seyfert galaxies] Extended=YES									
	Module					Subarray				
Dithers	B									
	FULL									
Spectral Elements	#	Primary Dither Type	Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE			STANDARD				4	
Special Requirements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	F200W	F444W	SHALLOW4	5	1	4	4	1030.73	
Special Requirements	Offset 40.0 arcsec, 40.0 arcsec									